KD Car

ME 100 Electronics for the Internet of Things

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1 Introduction

ME 100 aims to provide students with the skills to design and implement electronic systems that interact with their surroundings and communicate wirelessly. Additionally, it offers an overview of key concepts in Electrical Engineering, including analog and digital electronics, circuit design, and modern applications.

This project focuses on showcasing the skills mentioned above by the development and implementation of a remote-controlled (RC) car. The RC car serves as a practical application of key concepts in electrical engineering and IoT, bridging the gap between theoretical knowledge and hands-on experience. Using sensors, micro-controllers, and the ESP-NOW wireless communication protocol, the project demonstrates the integration of hardware and software to create a cohesive and interactive system.

This report details the design process, circuit architecture, IoT implementation, and performance evaluation of the RC car. The challenges encountered during the development and the innovative solutions applied are also discussed, focusing on the learning outcomes of the project. The work showcases how the principles of IoT and circuit design can be used to create intelligent and adaptable systems.

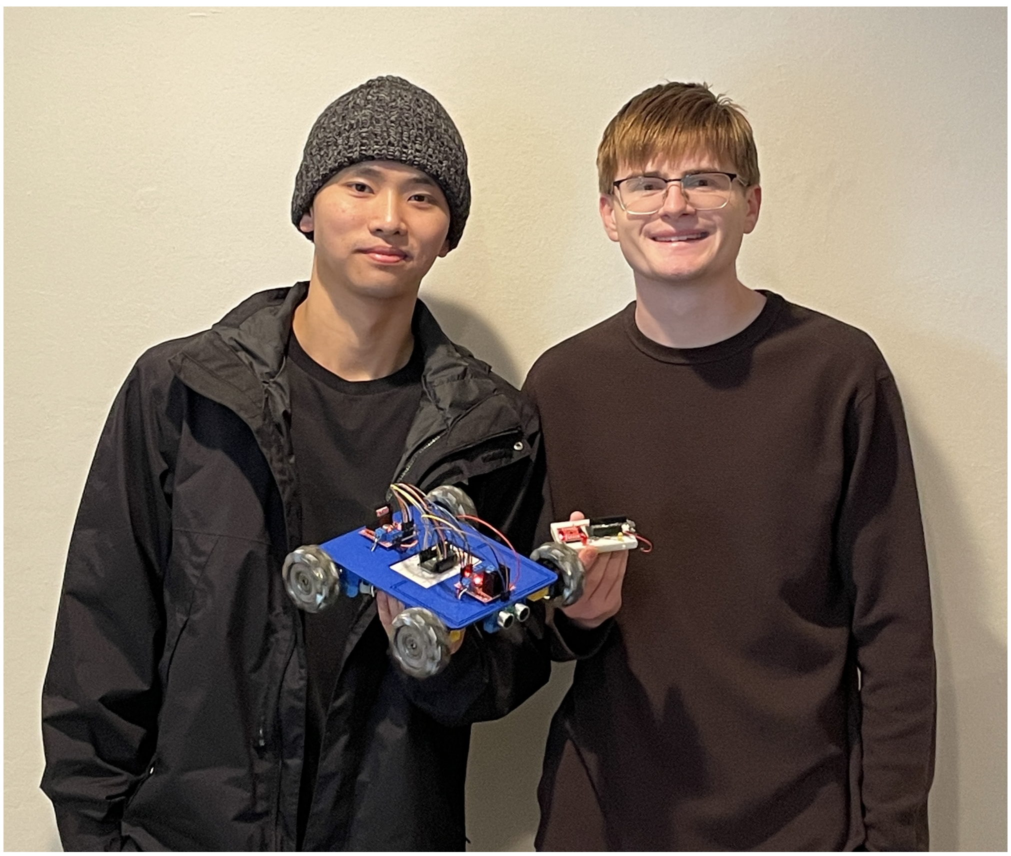

Figure 1: Huanbin and Drew holding the car and controller.

2 Hardware

As soon as we decided on creating an RC car, we had to decide on the car architecture. A holistic approach was taken to ensure that all hardware fit neatly together.

2.1 Wheels

At the outset of the project, we opted to use Mecanum wheels. This choice aligned with our primary project goals, which emphasized practical applications of circuitry and IoT rather than mechanical design. Mecanum wheels were particularly advantageous because they streamlined the mechanical design process. Furthermore, as illustrated in Figure 2, the unique mechanics of Mecanum wheels facilitate multidirectional movement through specific motor actuation combinations which presented a coding optimization challenge.

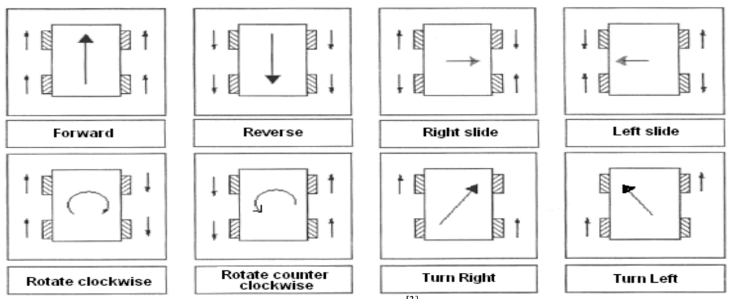

Figure 2: Mecanum Wheel Logic [1]

2.2 Motors

We selected four TT motors for their affordability and simplicity. These motors operate on a voltage range of 3–6 VDC, with higher voltages corresponding to increased rotational speeds, up to a maximum of 200 RPM. Although the motor speed can be finely adjusted using pulse width modulation (PWM), we opted for a binary control setup to reduce system complexity.

2.3 Motor Drivers

Two L298N motor drivers were employed to control the motors. Additionally, one of the motor drivers was utilized as a voltage regulator, enabling the ESP32 to be powered by the same 7.4V battery that supplies the motors.

2.4 Inertial Measurement Unit

The LSM6DSO was chosen as the Inertial Measurement Unit (IMU) for the project. It is connected to the ESP32 on the controller and measures acceleration and position along the X, Y, and Z axes. This data is transmitted to the ESP32 on the car via ESP-NOW, enabling precise control of motor actuation.

2.5 Ultra Sonic Sensor

The RCWL-1601 ultrasonic sensor, included in the lab kit, was used for obstacle detection. It was mounted on the front of the car, with data transmitted to the ESP32 on the controller. The system was programmed to stop all motors when the car detected an obstacle within 20 cm.

2.6 Light-Emitting Diodes

Two light-emitting diodes (LEDs) were integrated into the controller to provide a visual indication of the car's proximity to an obstacle. A yellow LED illuminated when an obstacle was detected between 60 cm and 30 cm, while a red LED activated when the obstacle was within 20 cm to 30 cm.

2.7 ESP-32

Two ESP32 microcontrollers were utilized to process sensor data and control the motors and LEDs. Communication between the two devices was established using the ESP-NOW wireless protocol, with both ESP32s functioning as "senders" and "receivers." The ESP32 on the car transmitted ultrasonic sensor data and received IMU data, while the ESP32 on the controller sent IMU data and received ultrasonic sensor data.

2.8 Wiring

2.8.1 Controller

The controller wiring is relatively straightforward. A red LED is connected to pin 14 on the ESP32, while a yellow LED is connected to pin 32, with each LED soldered to a 470-ohm resistor to limit current flow. The ESP32 powers the IMU through its 3.3V pin. Additionally, the serial data line (SDA) and serial clock line (SCL) of the ESP32 are connected to their respective pins on the IMU. Finally, all components share a common ground, ensuring proper operation.

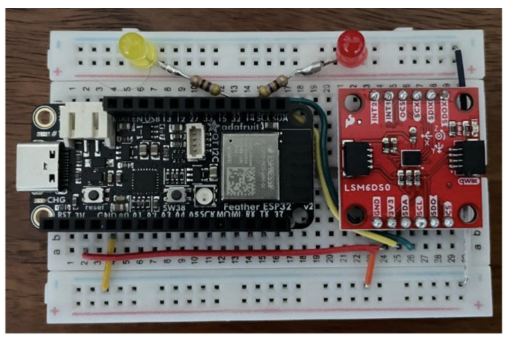

Figure 3: Controller Wiring

2.8.2 Car

The wiring schematic for the car is shown below. Each motor driver is connected to two motors via the OUT pins. The control and PWM pins of the motor drivers are linked to the ESP32. The battery is connected directly to the "back" motor driver, which in turn passes power to the front motor driver via an additional wire. The ESP32 receives 5V from the front motor driver. A common ground is shared to ensure proper functionality.

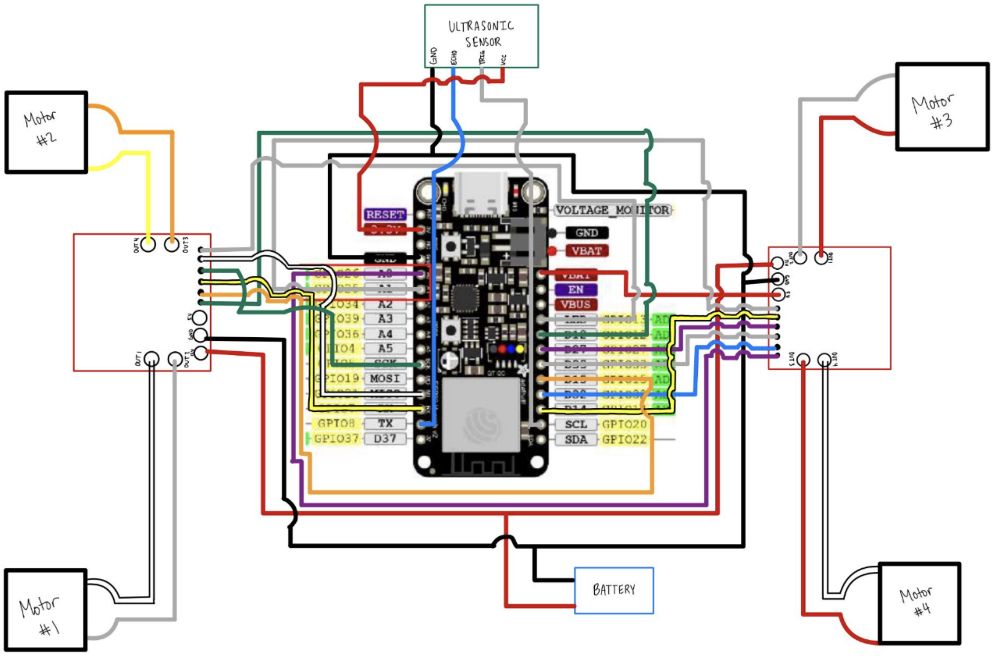

Figure 4: Car Wiring Diagram

3 Software

There are two different sets of code. One is for the ESP-32 on the controller, and the other is for the ESP-32 on the car.

3.1 Controller

The main features of the controller code include the initialization of ESP-NOW, accelerometer data transmission, ultrasonic data reception and processing, and LED control. The code begins by initializing the ESP-NOW wireless communication protocol to enable data exchange. A peer MAC address is configured to establish communication with the ESP32 on the car.

The LSM6DSO accelerometer is initialized to measure acceleration along the x, y, and z axes. Continuous monitoring of accelerometer data occurs, and significant movements exceeding a defined threshold (0.05) trigger the transmission of data to the peer device. The acceleration data is packaged into a JSON object and sent via ESP-NOW. If the accelerometer fails to initialize, the program halts execution to prevent further operation.

Incoming ultrasonic sensor data is received, decoded, and parsed to determine the distance. If the distance is between 20 and 30 units, the red LED (connected to pin 14) lights up, signaling the car to stop. If the distance falls between 30 and 60 units, the yellow LED (connected to pin 32) lights up, indicating a medium range. For all other distances, both LEDs remain off.

3.2 Car

The main features of the car code include ESP-NOW communication, distance measurement, and motor control.

The ESP-NOW protocol is initialized for peer-to-peer communication. The RC car communicates with the controller by sending ultrasonic sensor distance data and receiving IMU data.

An RCWL-1601 ultrasonic sensor is used to measure the distance from obstacles. If the distance is less than 20 cm, the motors are stopped for safety. Distance data is sent to the controller so that the LEDs light up at appropriate times.

Four motors are controlled using PWM signals for speed (although there are only two different speed settings) and GPIO pins for direction. Functions are defined to stop all motors or control them based on received commands. The control motors function processes incoming data from the IMU (x, y, z values) to determine movement. Different movements include forward and backward, sliding left or right, and diagonal left or right movement. Speed is set to the maximum (1023).

4 Project Outcome

4.1 Project Reflection

The project is considered to be successful, but it is also far from perfect.

The two main causes are the physical limitation of the sensors(hardware limitation) and the controller code (software limitation). The ultrasonic sensor is not the most suitable option to achieve the full functionality of this project due to its limited accuracy and susceptibility to interference from the surrounding environment. Therefore, choosing a more optimal sensor will fundamentally avoid imprecise computation.

On the software side, improvements such as optimizing the code to reduce its complexity can lead to faster computations and help address latency issues.

Overall, the project provided a valuable hands-on opportunity to apply key concepts from the course, such as troubleshooting circuits, ideas of IoT, and implementing logical control, within a practical design context. It facilitated a deeper understanding of the development process for creating a functional project using fundamental electronic elements, including micro-controllers, sensors, and actuators. Through this experience, the integration of theoretical knowledge with practical skills was reinforced, strengthening both technical competence and problem-solving abilities.

4.2 Future Improvement

To enhance the project in the future, several improvements can be implemented on both the hardware and software sides.

On the hardware side, replacing the ultrasonic sensor with a LiDAR sensor will significantly improve robustness and functionality because it outputs higher precision data and is less affected by the vibrations of the car. The wiring on the top side of the car can also be made more neat.

On the software side, other than optimizing the controller code, employing a real-time operating system will enhance task scheduling and resource management within ESP32 which ensures a smoother response between sender and receiver.

4.3 Demo Video Link

The link to the demo video is: `https://youtube.com/shorts/TiWNzglT_nE?feature=share`

Nomenclature

IMU	Inertial Measurement Unit
IoT	Internet of Things
LED	Light Emitting Diode
PWM	Pulse Width Modulation
RC	Remote Controlled
RPM	Rotations Per Minute
SCL	Serial Clock Line
SDA	Serial Data Line
TT	Tourist Trophy
V	Volts
VDC	Volts Direct Current

References

[1] J. Phillips, *Mechatronic design and construction of an intelligent mobile robot for educational purposes.* PhD thesis, Massey University, Palmerston North, New Zealand, 2000.